

MONITORING AND MANAGEMENT SYSTEM OF RESIDENTIAL ELECTRICITY CONSUMPTION

INSTALLATION

This document presents the steps to follow for the installation of the system.

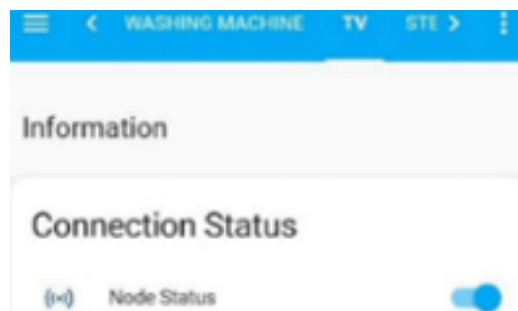
The system has two nodes. Node 1 for the monitoring and management of household appliances and Node 2, for the monitoring of residential electricity consumption and circuit management.

Installation of Node 1

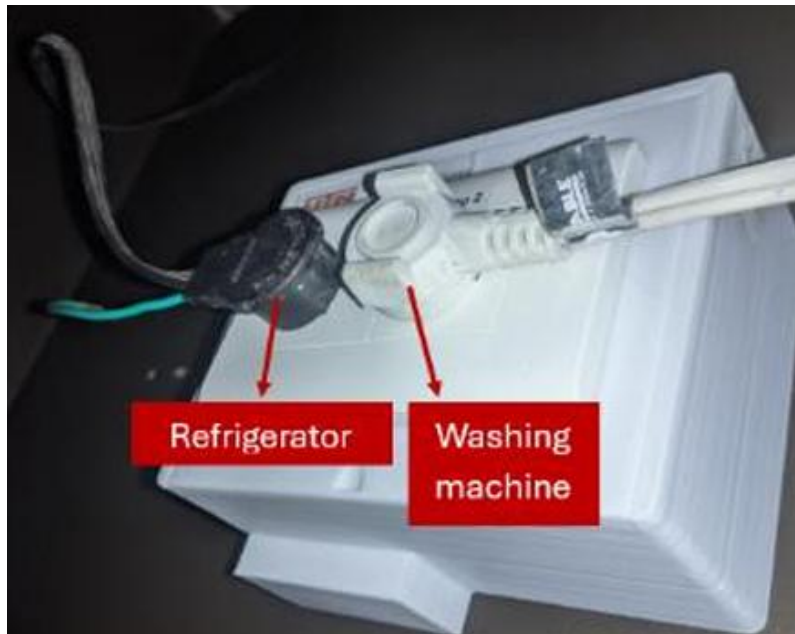
1. Node 1 is installed in any household outlet. It is recommended that it be in the most fenced to the appliances that are going to be monitored.
2. Turn on the node via the on/off button.



3. On the Home Assistat platform check the Active Status of the Node.

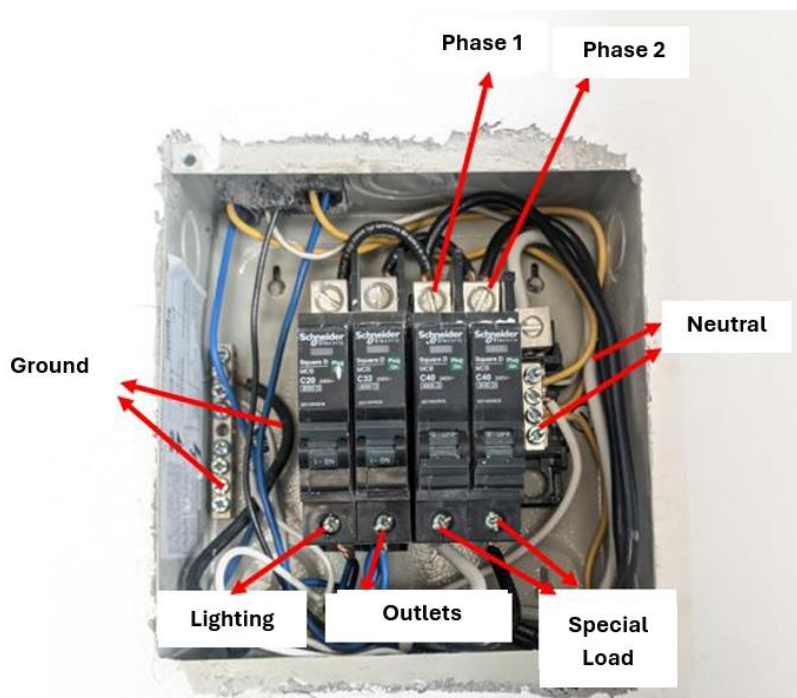


4. On the outlets of the node, connect the appliances to be monitored

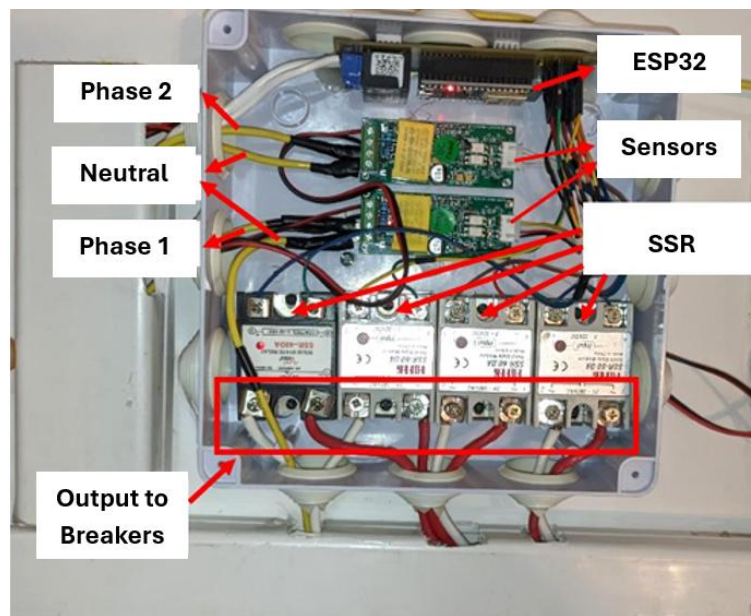


Installation of Node 2

1. Identify the conductors designated as Phase (1 and 2), the Neutral and Ground bars or terminals, on the electrical distribution board.
2. Identify the electrical circuits present. For example lighting, outlets and special loads.



3. Relay the connections from the Phase and Neutral terminal block of the distribution board the PZEM004-T sensors, for Phase 1 and Phase 2 voltage measurement.
4. The current transformers (CT) are placed through the conductors designated as Phase 1 and 2, for the current measurement each Phase.
5. Connect the solid-state relays (SSR) individually to each output of the breakers of the electric circuits, to manage the loads by means of activation or deactivation from the ESP32 microcontroller.



6. Close and tighten the screws of the top cover of the node, to avoid any electrical risk.

